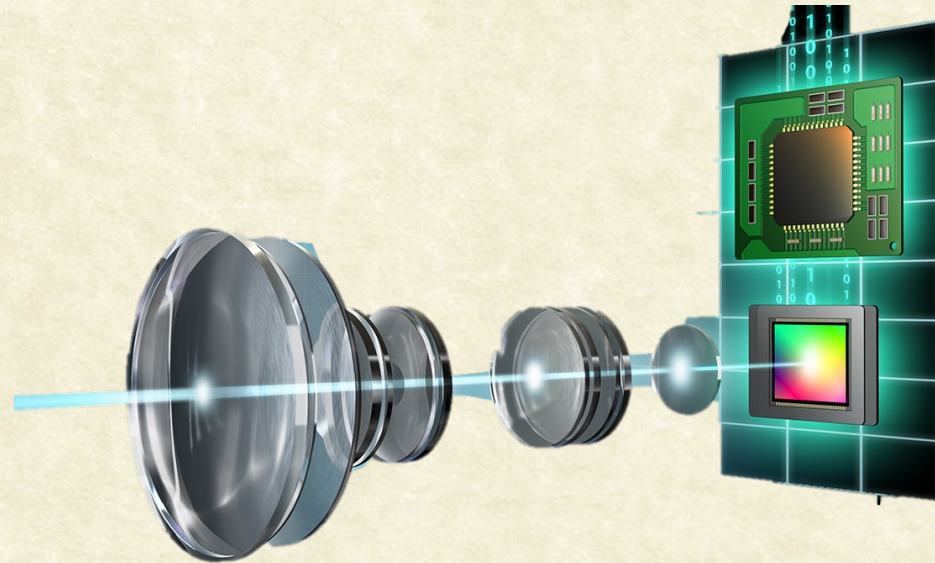




CS7.404: Digital Image Processing

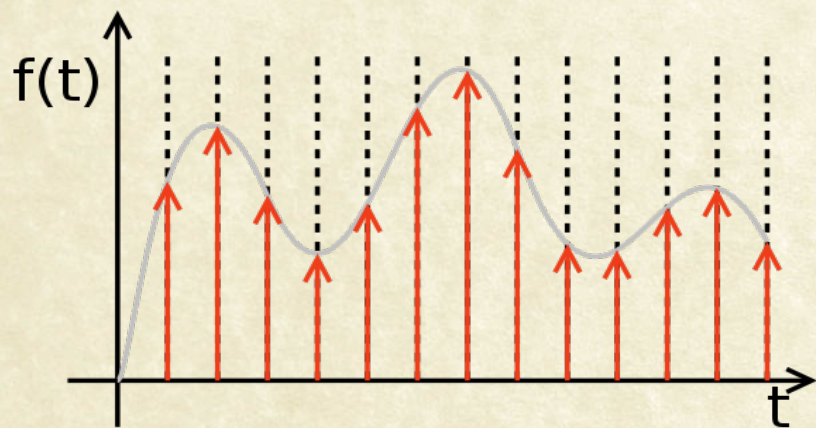
Monsoon 2025: 0: Sampling and Quantization (Recap)



Anoop M. Namboodiri
Biometrics and Secure ID Lab, CVIT,
IIIT Hyderabad



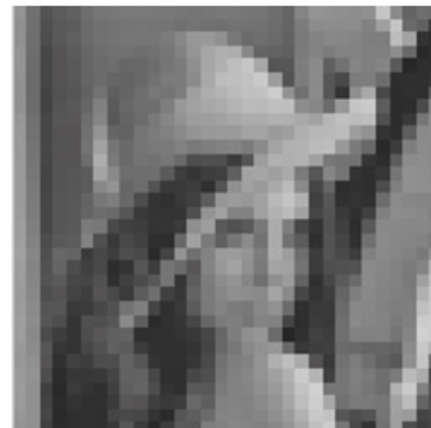
Summary



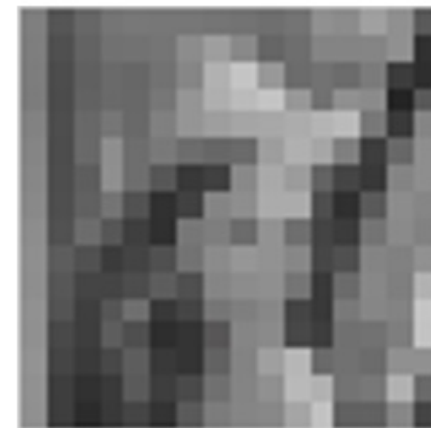
Sampling



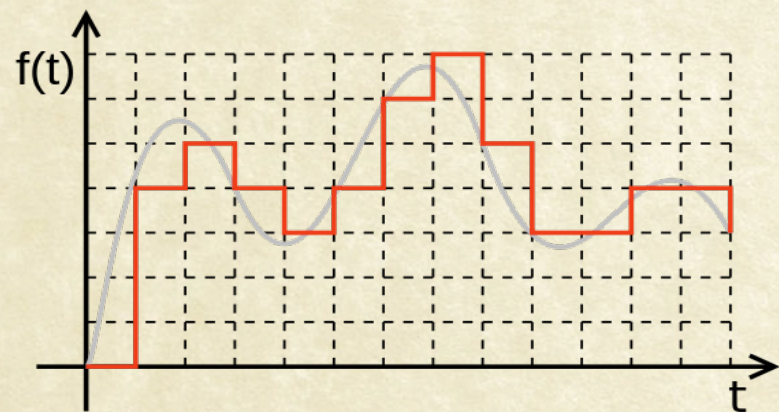
256 × 256



32 × 32



16 × 16



Quantization



8 bits per pixel



4 bits per pixel



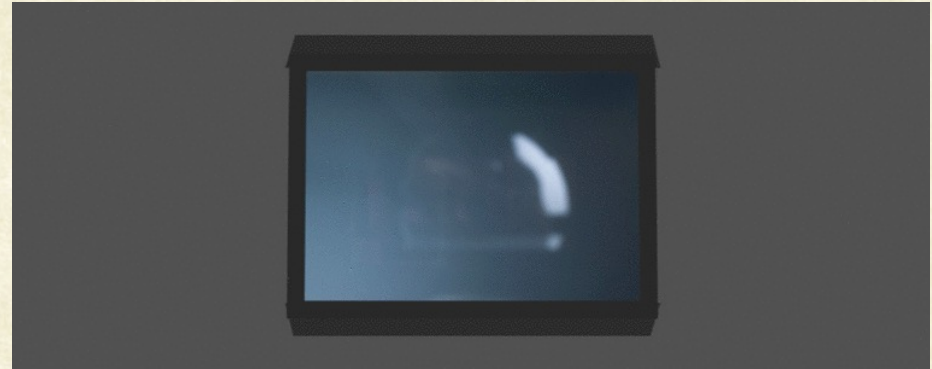
2 bits per pixel

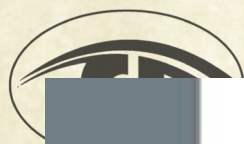


1 bit per pixel



Temporal Sampling





Aperture

small
aperture



F32



F22



F16



F11



F8



F5,6



F4



F2,8



F2



F1,4

large
aperture

Shutter

fast
shutter speed



1/1000



1/500



1/250



1/125



1/60



1/30



1/15



1/8



1/4



1/2

slow
shutter speed

ISO

low
sensitivity



ISO 50



ISO 100



ISO 200



ISO 400



ISO 800



ISO 1600



ISO 3200



ISO 6400



ISO 12800



ISO 25600

high
sensitivity



Effect of Shutter Speed





Temporal Sampling





Topics for Today

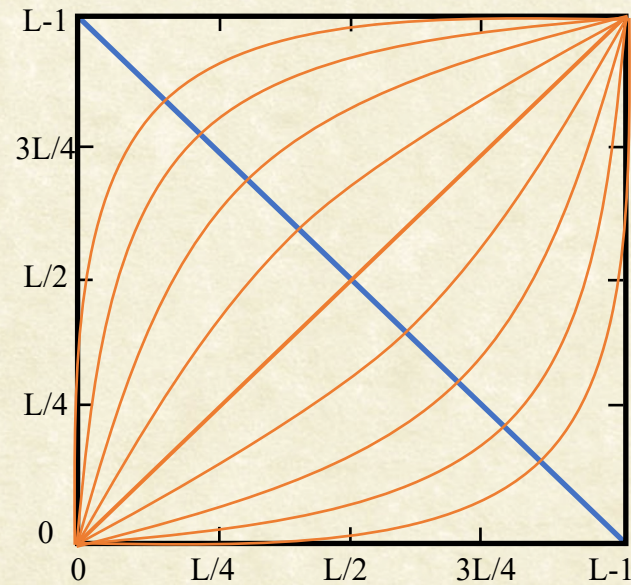
Pixel Processing (Sections 3.1 – 3.3)

- I. Linear Transformations
- II. Non-Linear Transformations
- III. Histogram Manipulation



CS7.404: Digital Image Processing

Monsoon 2025: I: Pixel Processing (Linear Transformations)



Anoop M. Namboodiri
Biometrics and Secure ID Lab, CVIT,
IIIT Hyderabad



Two Paradigms of Image Processing

Directly in Spatial Domain

- Manipulating Pixels
 - Easier to Understand
 - Easier to Implement
 - Local Changes
 - Not so efficient

Processing in Transform Domain

- Manipulating JPG coefficients
 - Changes can be global
 - Can be very efficient, effective, ...
 - Complex to visualize
 - Tricky to implement



Intensity Transformations: Point-to-Point

$$b(x, y) = T(a(x, y)); \quad T(z) = L - z$$

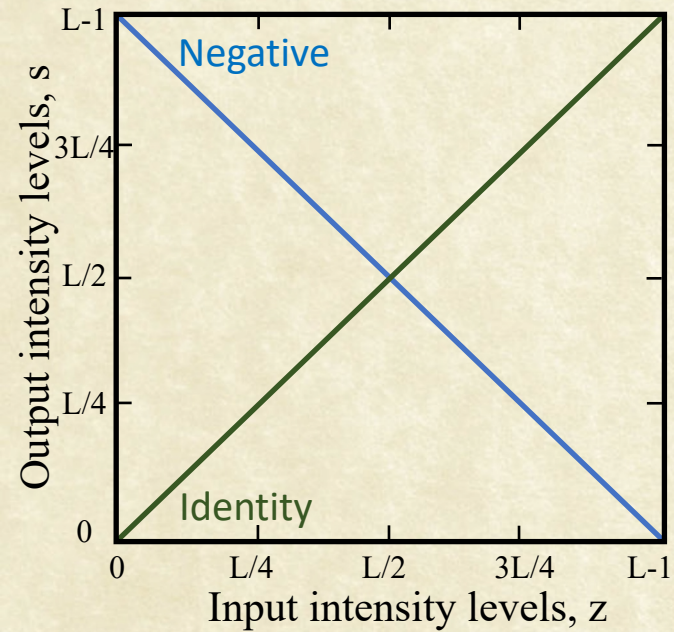
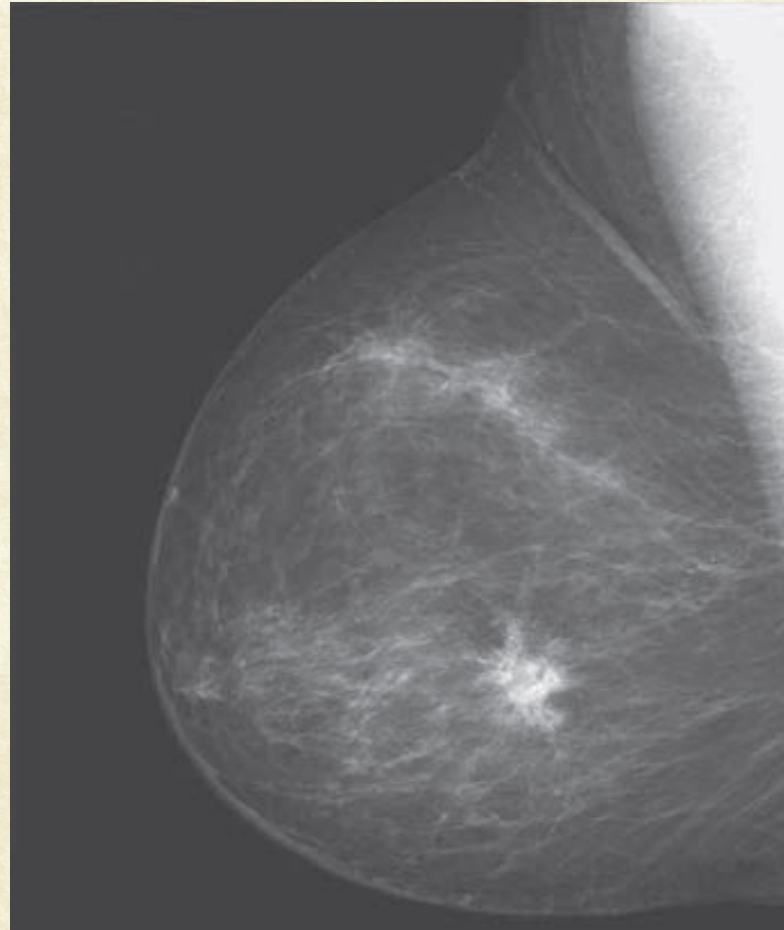


FIGURE 3.4

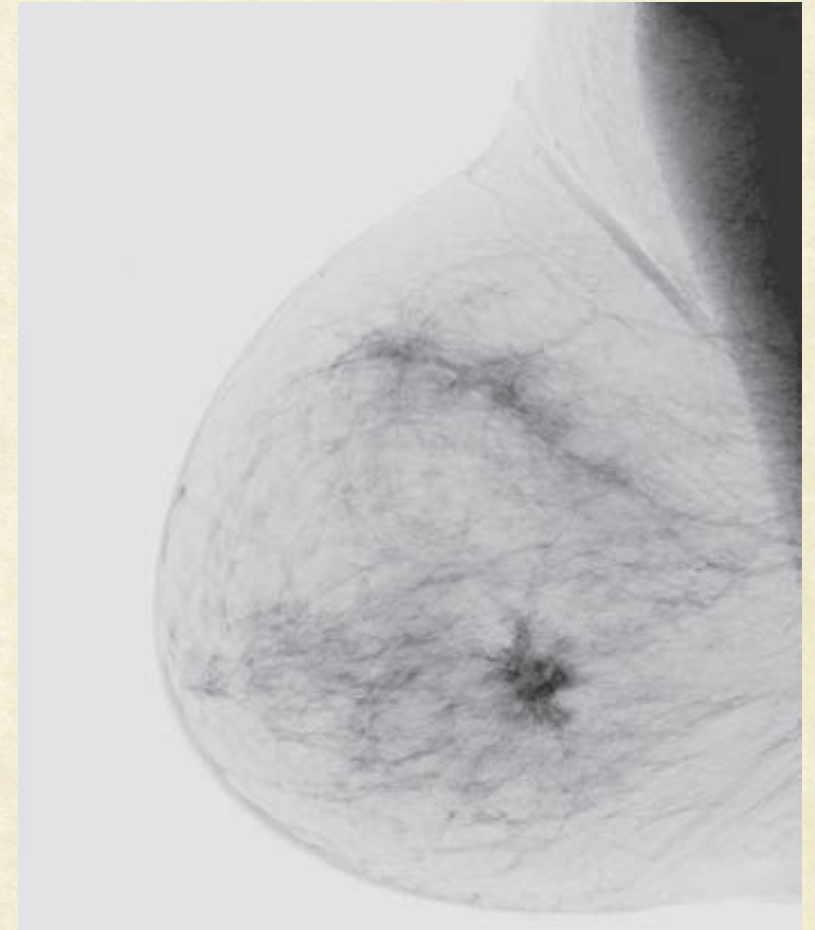
(a) A digital mammogram.

(b) Negative image obtained using the above equation.

(Image (a) Courtesy of GE Medical Systems.)



(a)

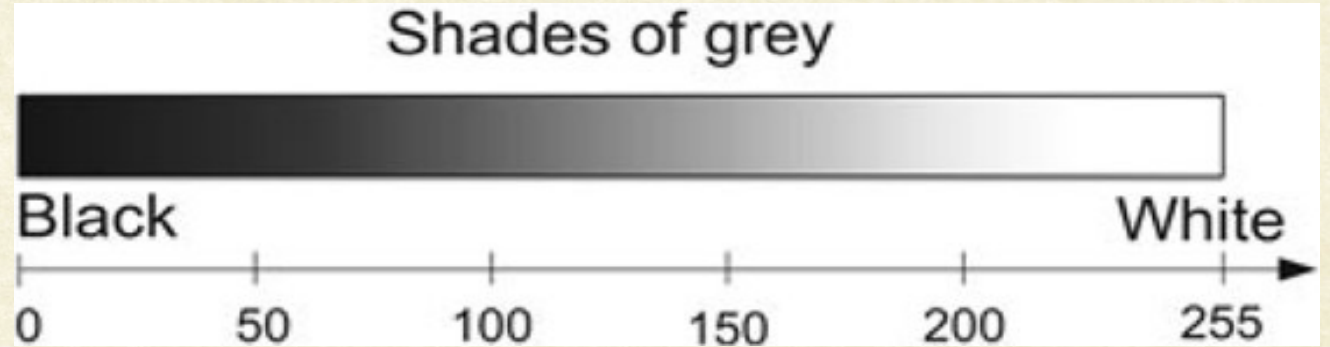


(b)



Linear Intensity Transforms

- $T(z) = z + k$
- $T(z) = z - k$
- $T(z) = kz$
- $T(z) = k_1z + k_2$



e.g.,

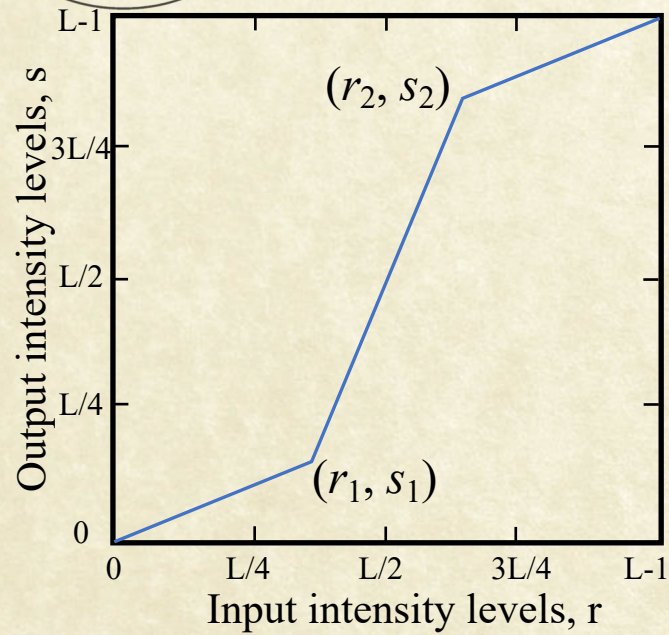
- Image Negative
- Data Visualization:
 - Map to display range

$$J = \text{round} \left(255 * \frac{I - \min(I)}{\max(I) - \min(I)} \right)$$

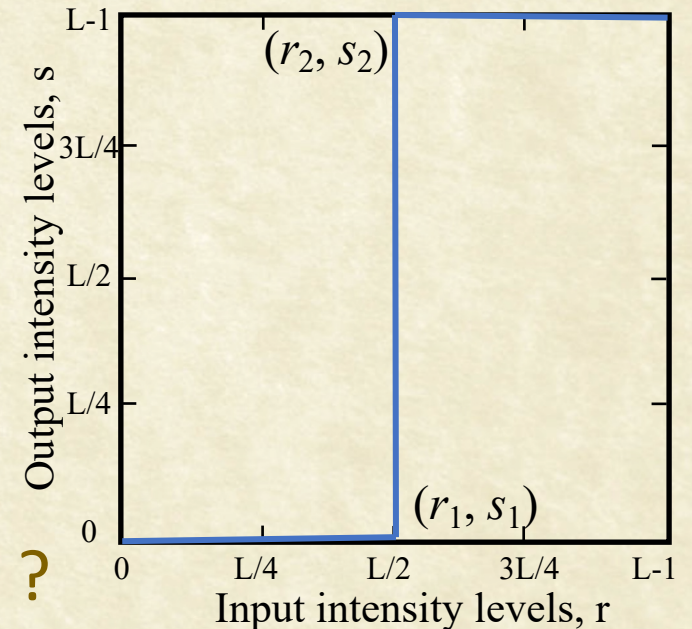
Qn: What do the intensity maps look like?



Piece-wise Linear Transformations



- Can be arbitrarily complex
- Finer control over transformation
- Thresholding
- Contrast Stretching
 - Expand intensity range to **full intensity range**
 - Qn: What are the constraints on (r_1, s_1) and (r_2, s_2) ?



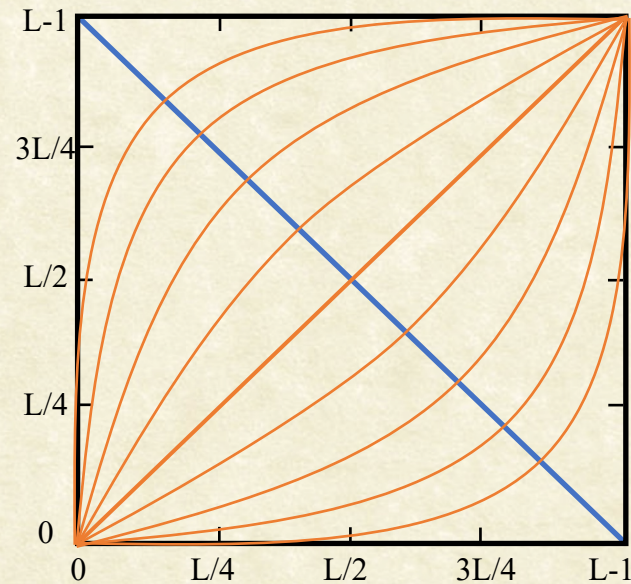


Questions?



CS7.404: Digital Image Processing

Monsoon 2025: II: Pixel Processing (Non-Linear Transformations)



Anoop M. Namboodiri
Biometrics and Secure ID Lab, CVIT,
IIIT Hyderabad



Log Transformations



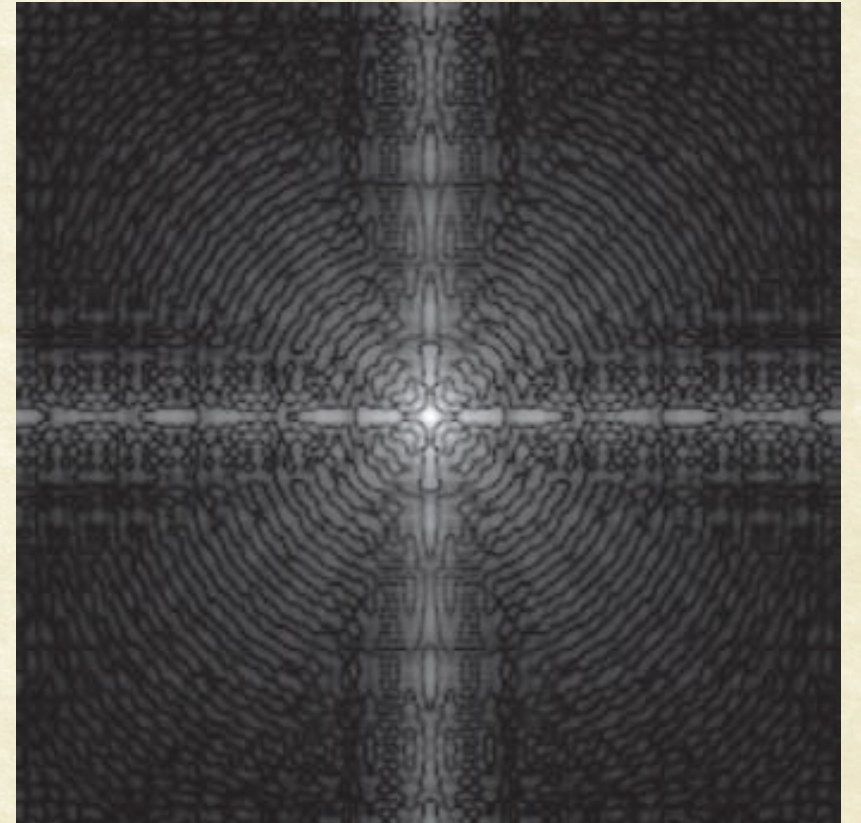
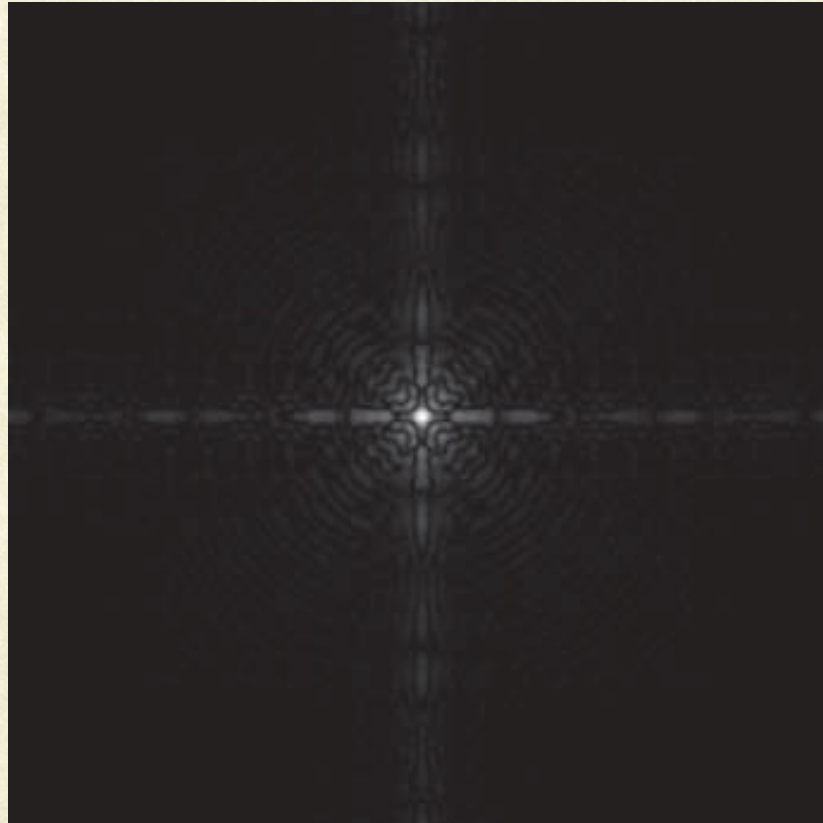
$$s = T(r) = c \log(1 + r)$$

FIGURE 3.5

(a) Fourier spectrum displayed as a grayscale image.

(b) Result of applying the log transformation in eq. with $c = 1$.

(Both images are scaled to the range $[0, 255]$.)





Power-Law Transformations

$$s = cr^\gamma$$

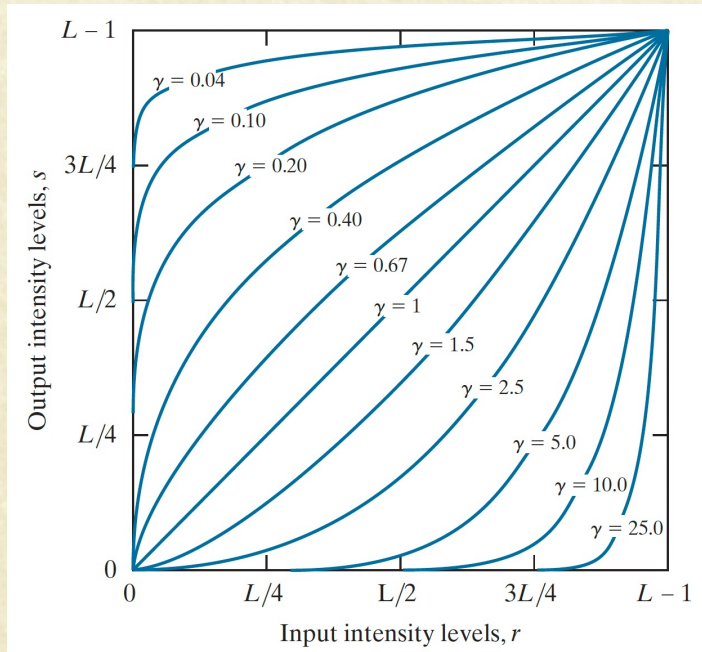


FIGURE 3.9

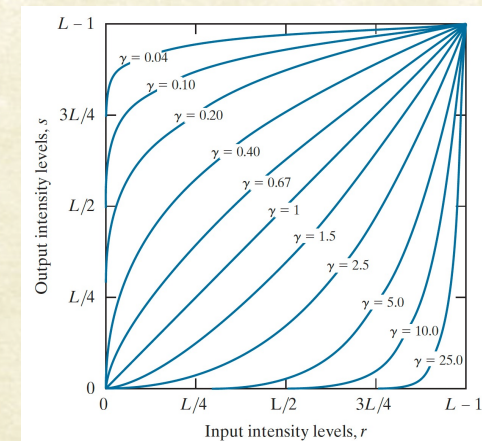
(a) Aerial image; (b)–(d) Results of applying the transformation in Eq. with $\gamma = 3.0, 4.0$, and 5.0 , respectively. ($c = 1$ in all cases.)
(Original image courtesy of NASA.)





Power-Law Transformations

$$s = cr^\gamma$$



$\gamma = 1$



$\gamma = 0.6$



$\gamma = 0.4$



$\gamma = 0.3$



Gamma Correction

FIGURE 3.7

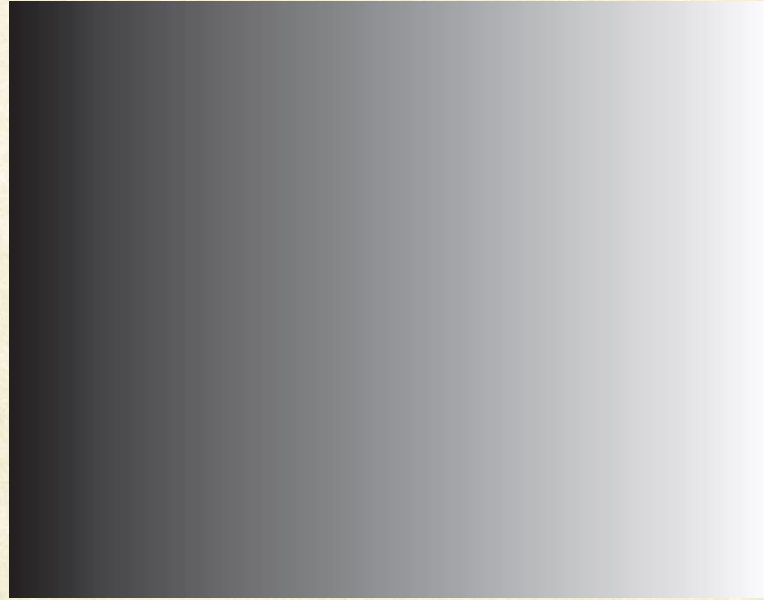
(a) Intensity ramp image.

(b) Image as viewed on a monitor with $\gamma = 2.5$.

(c) Gamma-corrected image.

(d) Corrected image as viewed on the same monitor.

Compare (d) and (a).



(a)



(b)



(d)



(c)



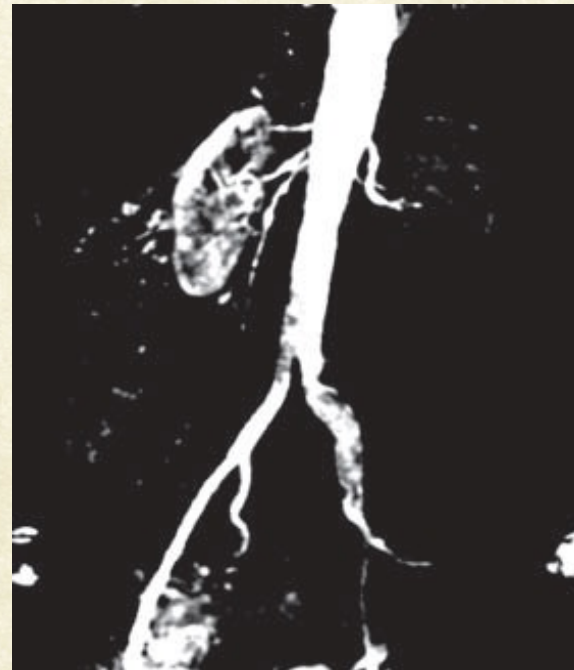
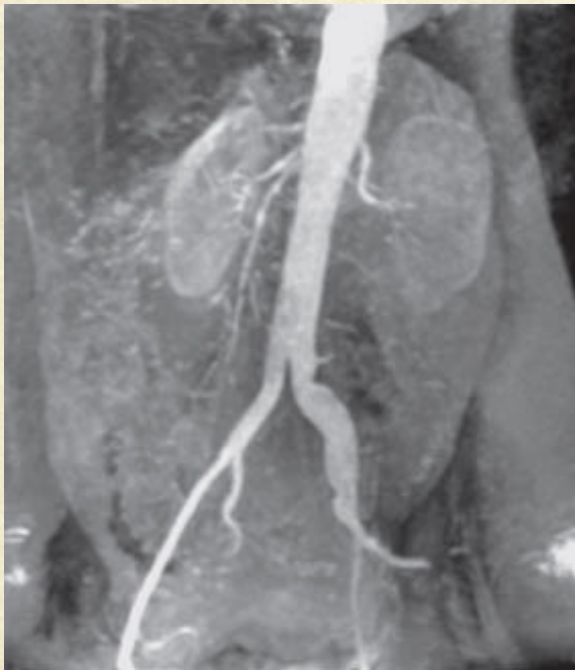
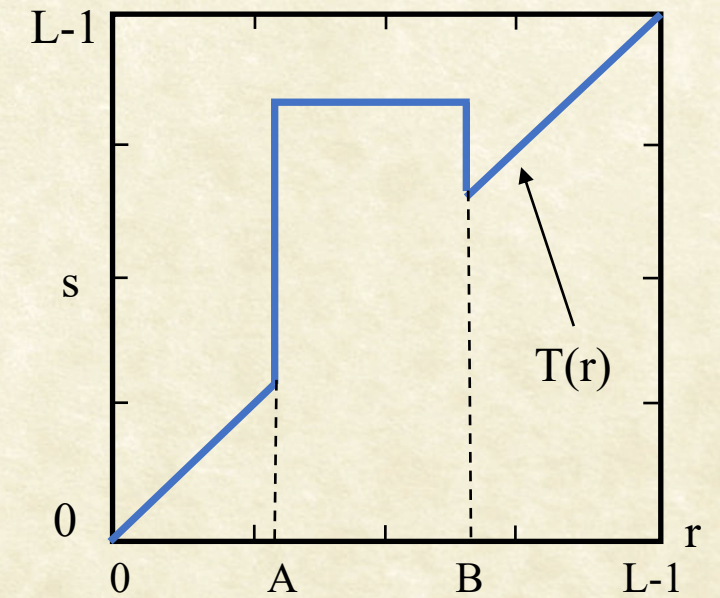
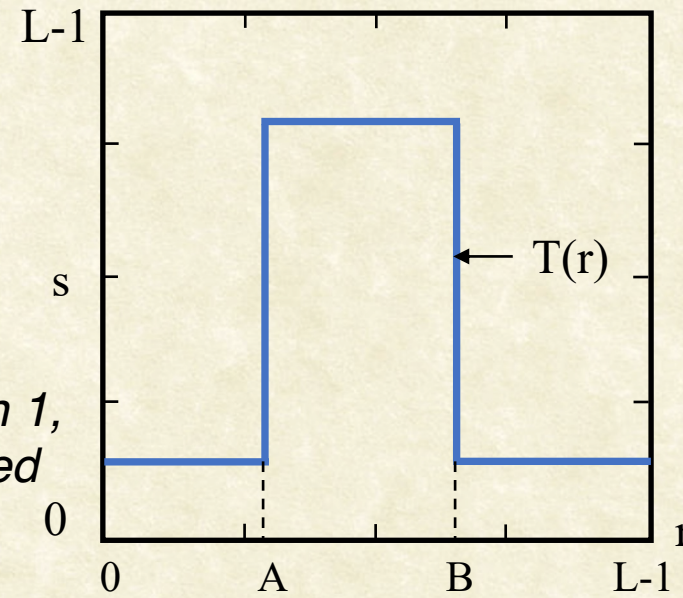
Intensity Slicing

FIGURE 3.12

(a) Aortic angiogram.

(b) Result of using a slicing transformation 1, with the range of intensities of interest selected in the upper end of the gray scale.

(c) Result of using the transformation 2





Bit plane slicing

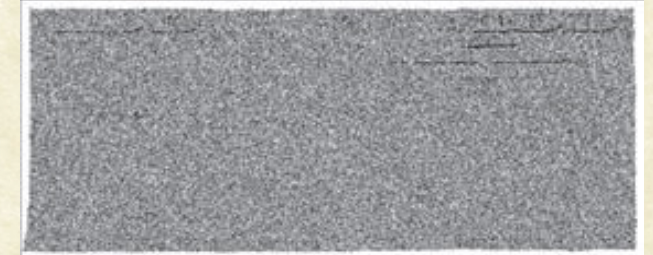
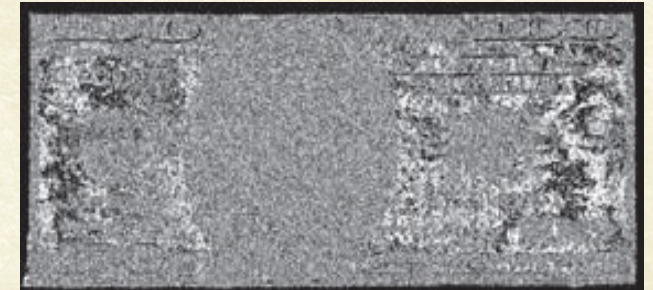
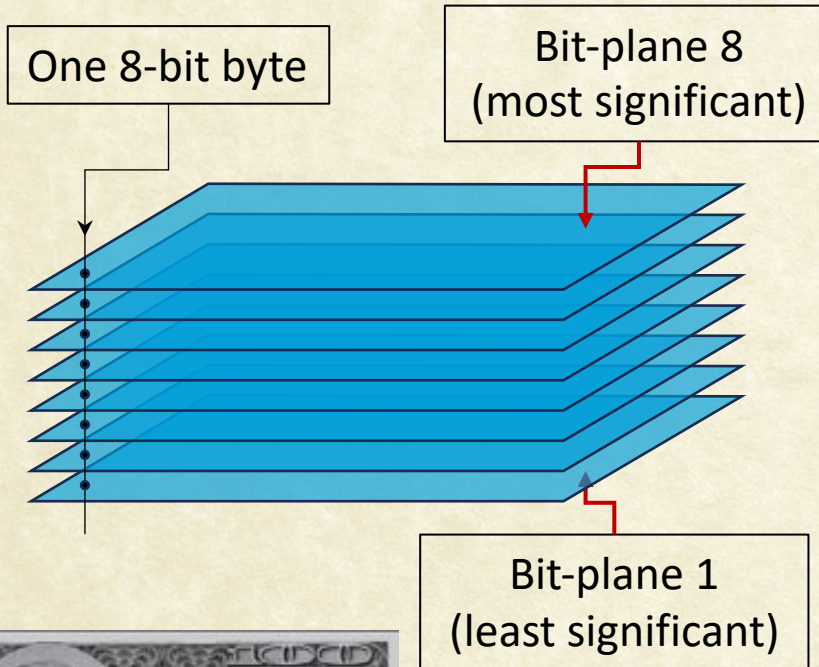


FIGURE 3.14 (a) An 8-bit gray-scale image of size 550 x 1192 pixels. (b) through (i) Bit planes 8 through 1, with bit plane 1 corresponding to the least significant bit. Each bit plane is a binary image.



Bit plane slicing

FIGURE 3.15 Image reconstructed from bit planes:
(a) 8 and 7
(b) 8, 7, and 6
(c) 8, 7, 6, and 5.



Original Image



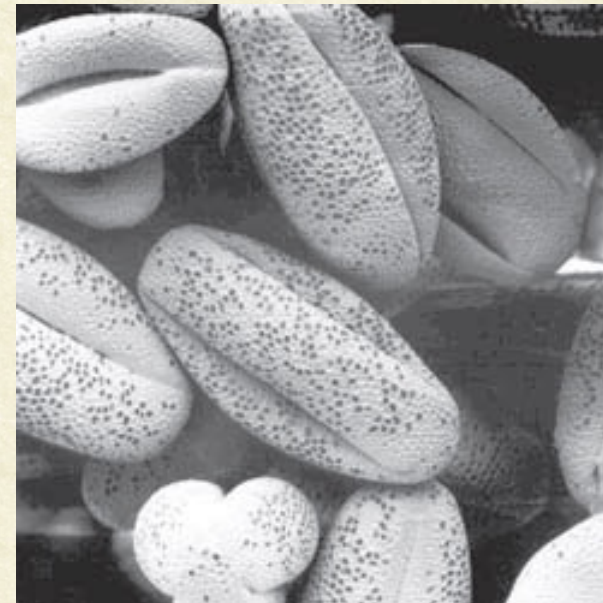
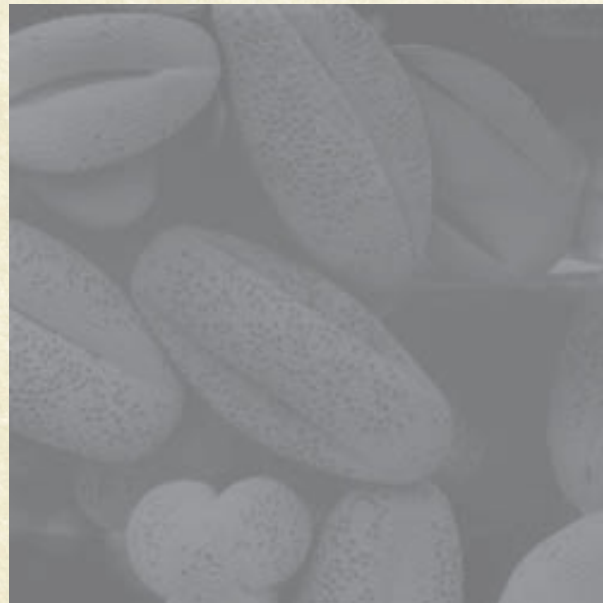


Questions?



CS7.404: Digital Image Processing

Monsoon 2025: III: Pixel Processing (Histogram Manipulation)



Anoop M. Namboodiri
Biometrics and Secure ID Lab, CVIT,
IIIT Hyderabad

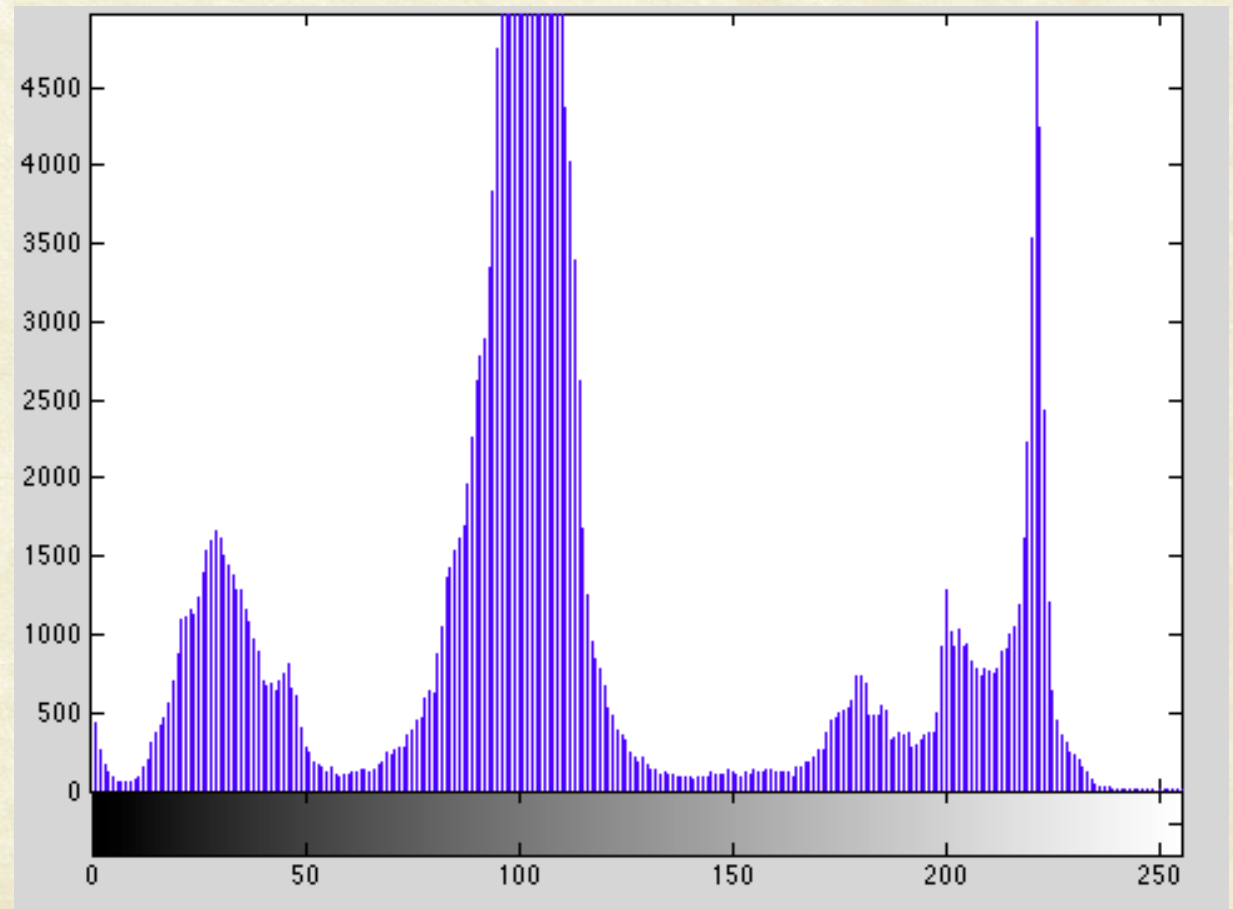


Histogram

$$h_r(i) = n_i$$

$i \rightarrow$ intensity value, range $[0, L-1]$

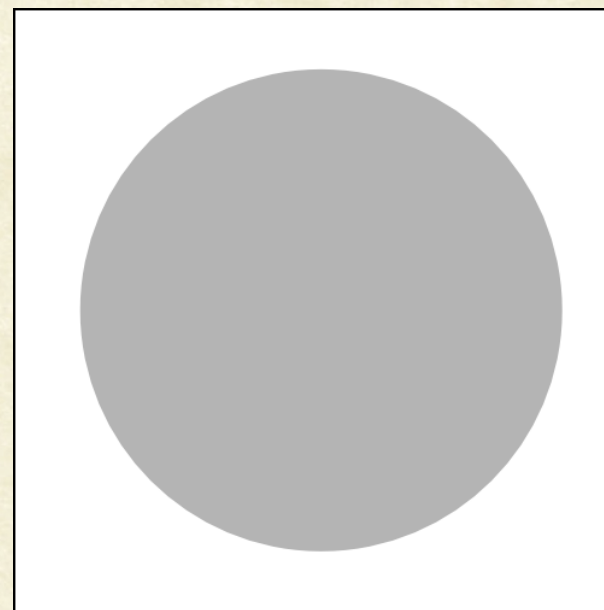
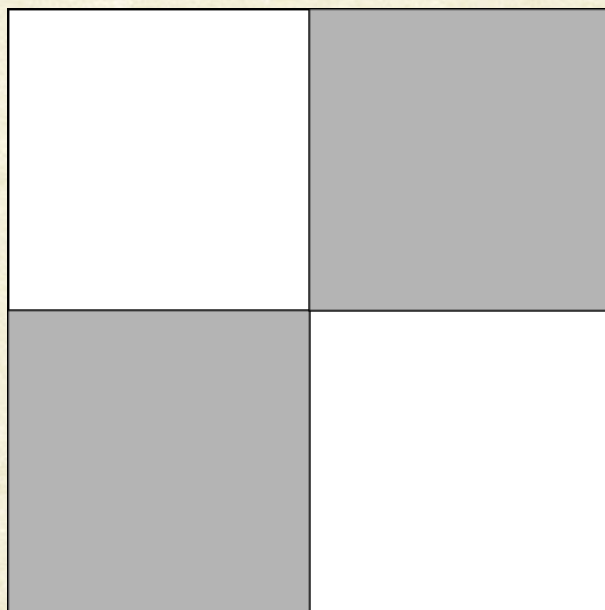
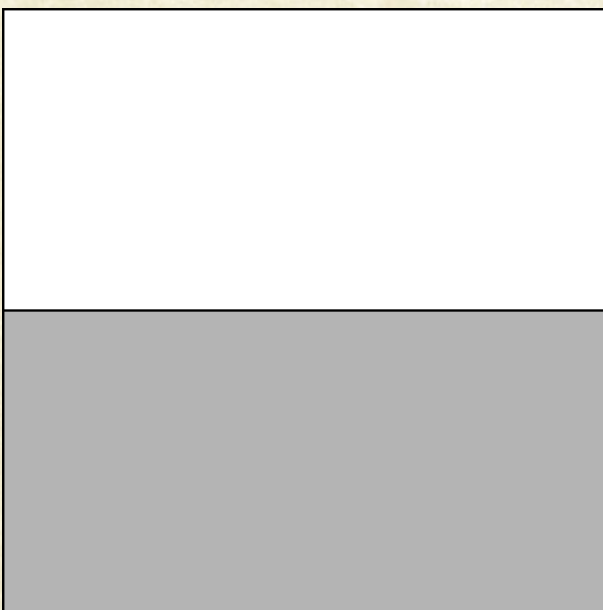
$n_i \rightarrow$ number of pixels with intensity i





Histograms

- ▶ Different images can have same histogram



- ▶ No information about spatial distribution of intensity values



Histograms

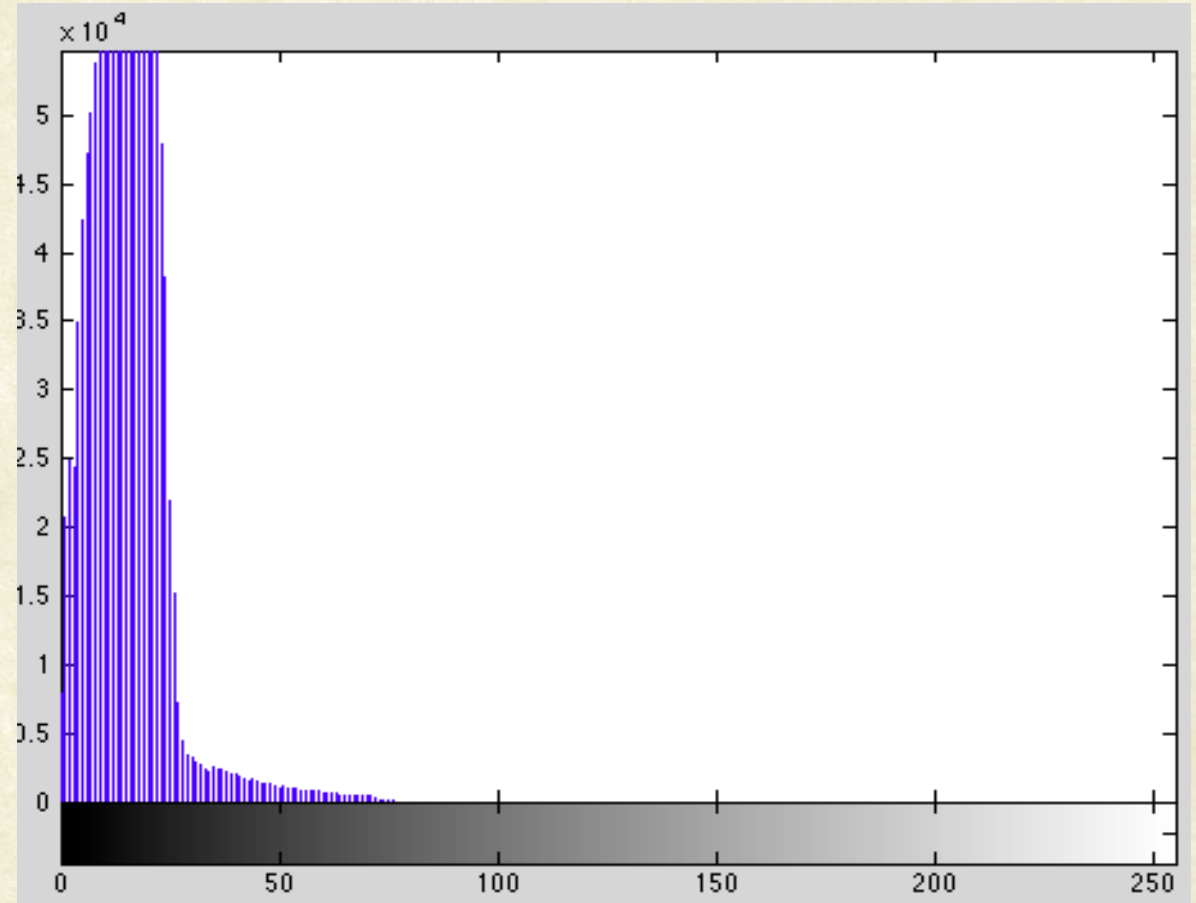
- ▶ What can we infer from histograms?



Histogram viewing standard in most DSLR cameras



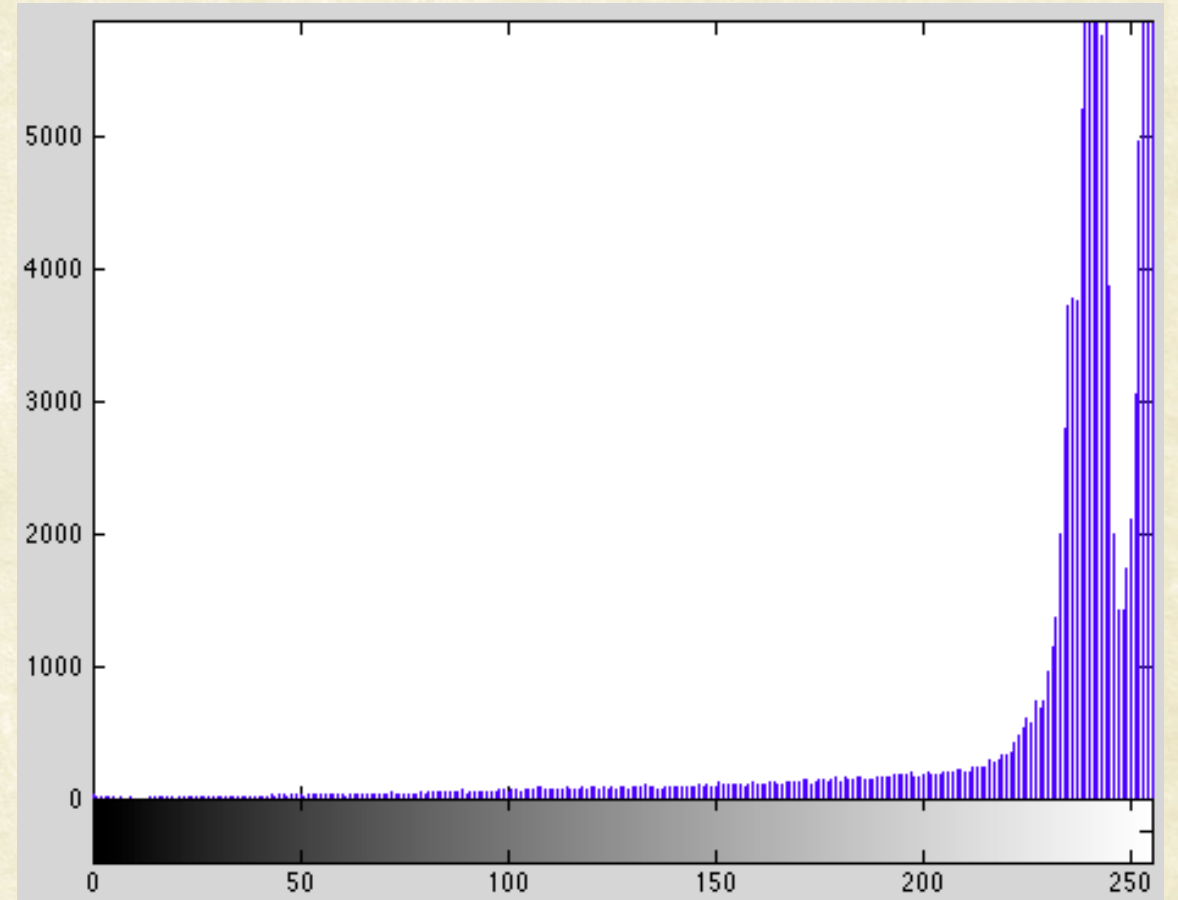
Histograms and Brightness



Under exposure



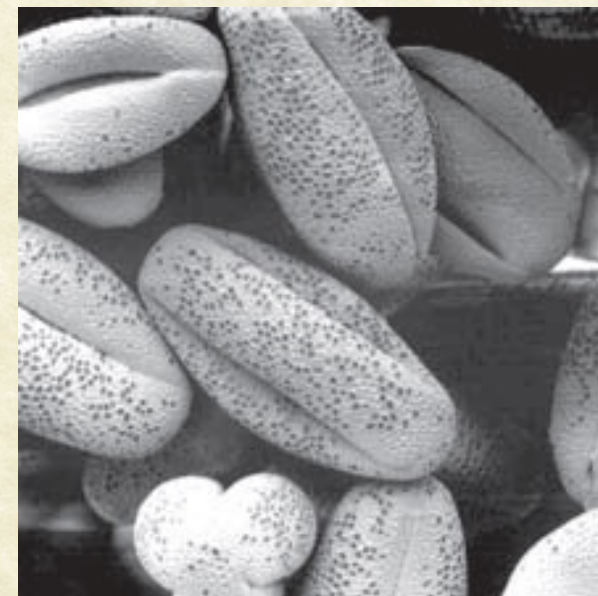
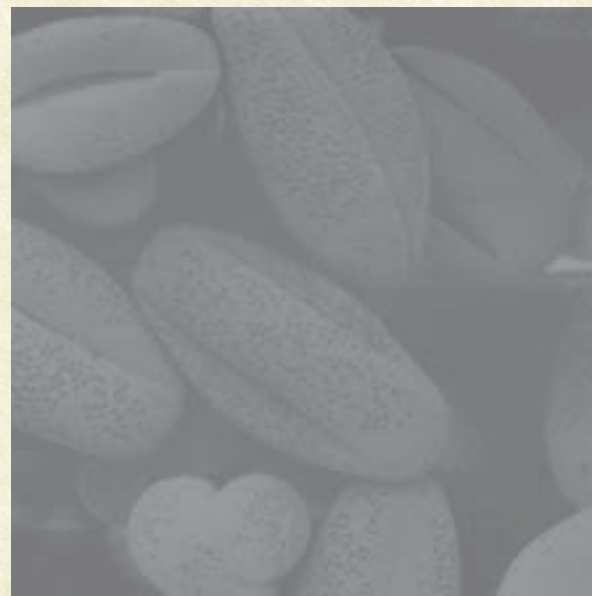
Histograms and Brightness



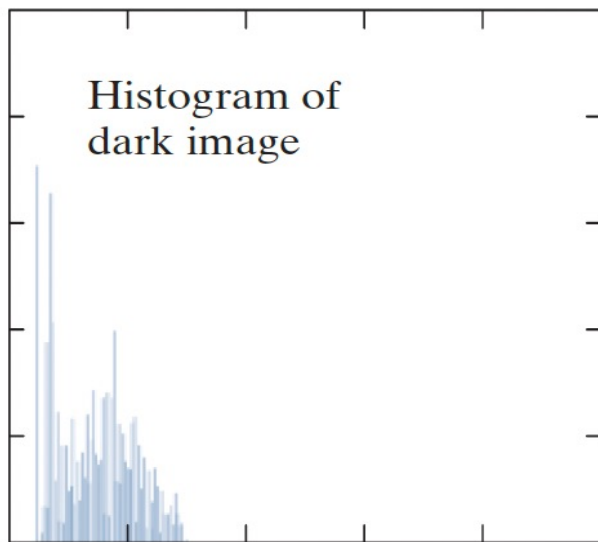
Over exposure



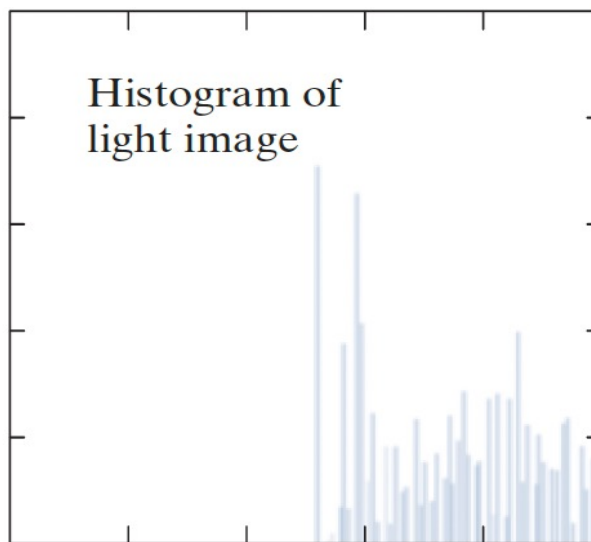
Histogram and Contrast



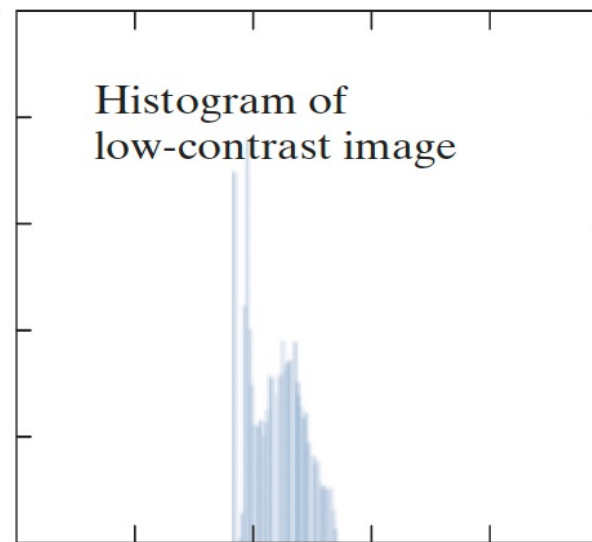
Histogram of
dark image



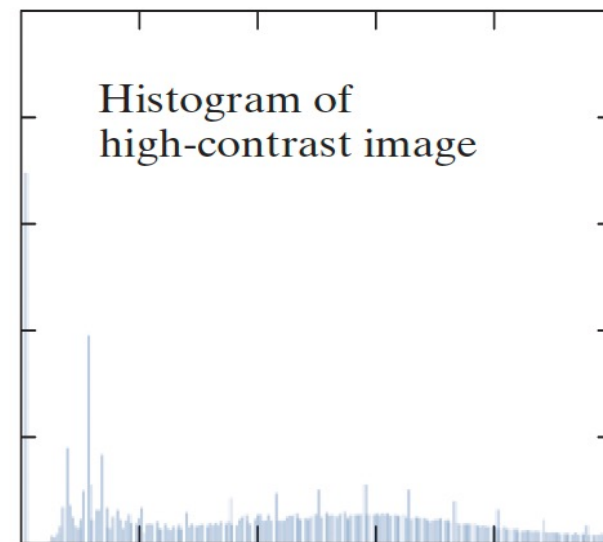
Histogram of
light image



Histogram of
low-contrast image



Histogram of
high-contrast image

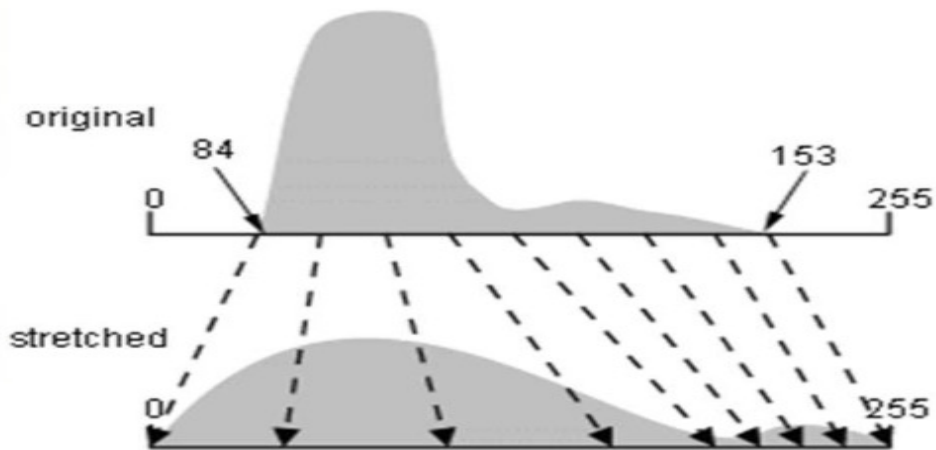


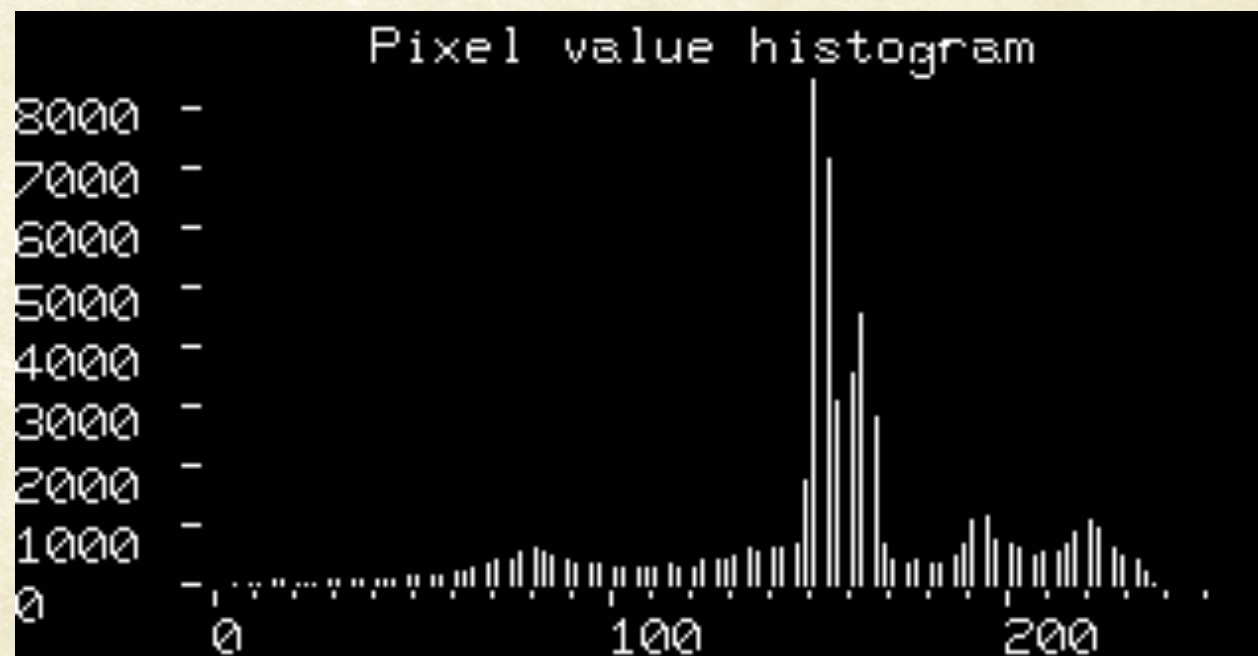
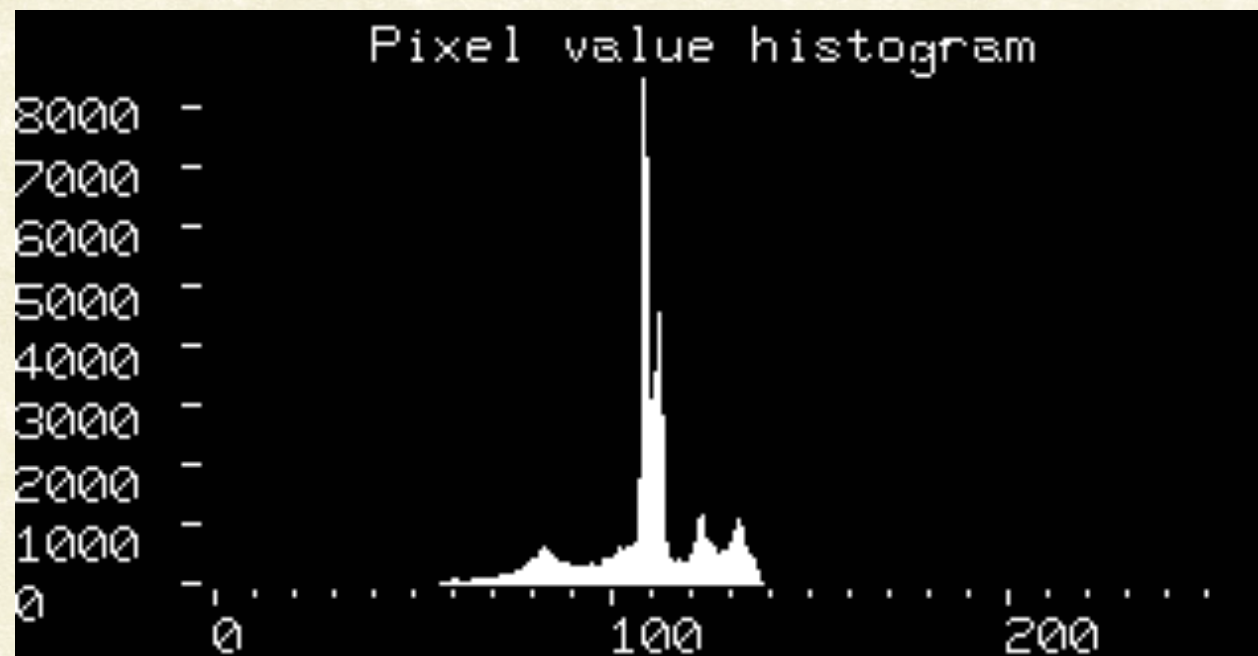


Contrast Stretching



$$f_{\text{ac}}(a) = a_{\text{min}} + (a - a_{\text{low}}) \cdot \frac{a_{\text{max}} - a_{\text{min}}}{a_{\text{high}} - a_{\text{low}}}$$







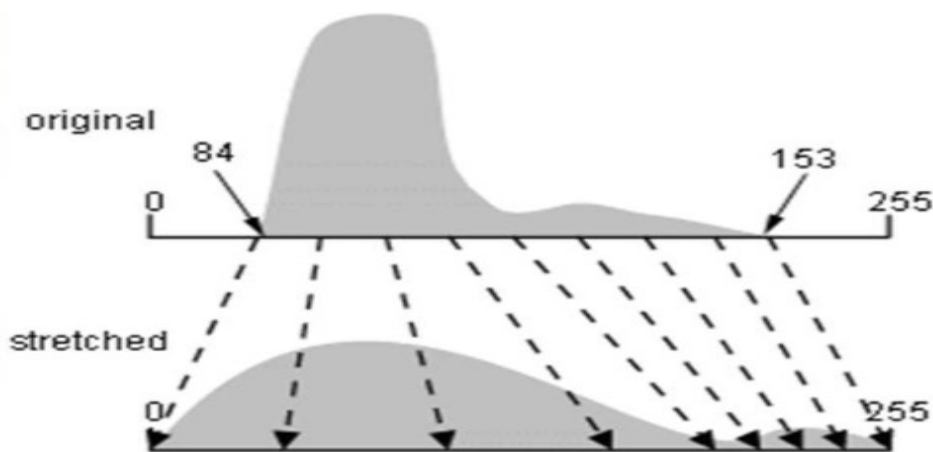
Contrast Stretching



$$f_{\text{ac}}(a) = a_{\text{min}} + (a - a_{\text{low}}) \cdot \frac{a_{\text{max}} - a_{\text{min}}}{a_{\text{high}} - a_{\text{low}}}$$

If $a_{\text{min}} = 0$ and $a_{\text{max}} = 255$

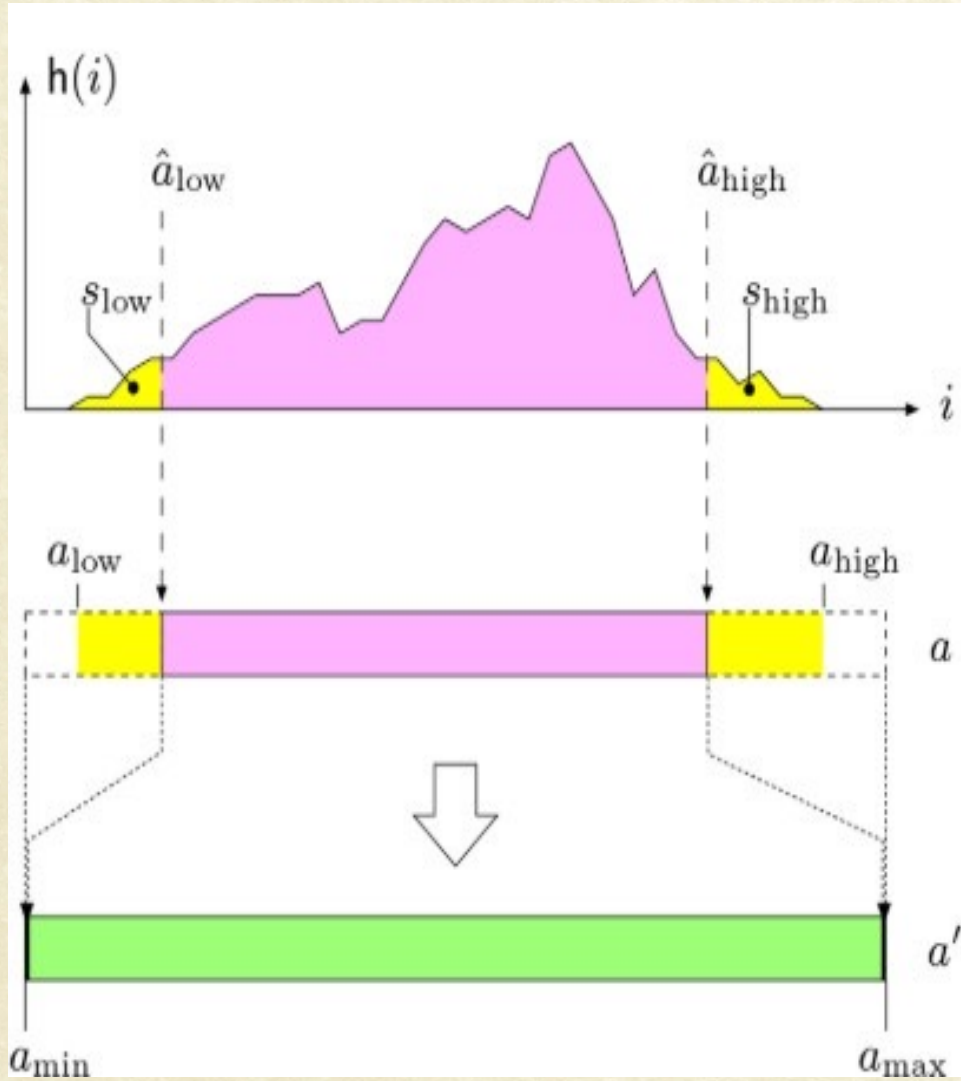
$$f_{\text{ac}}(a) = (a - a_{\text{low}}) \cdot \frac{255}{a_{\text{high}} - a_{\text{low}}}$$



What happens if we have a single pixel with intensity 255 or 0 in the original intensity range?



Contrast Stretching: Ver.-2



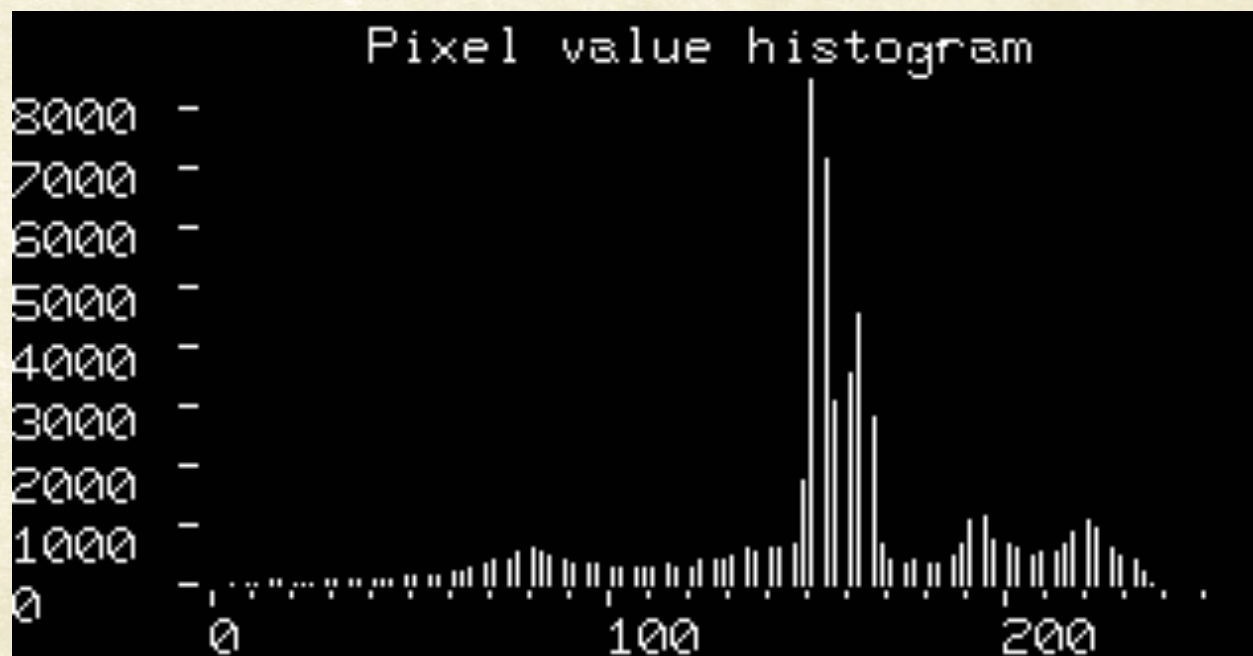
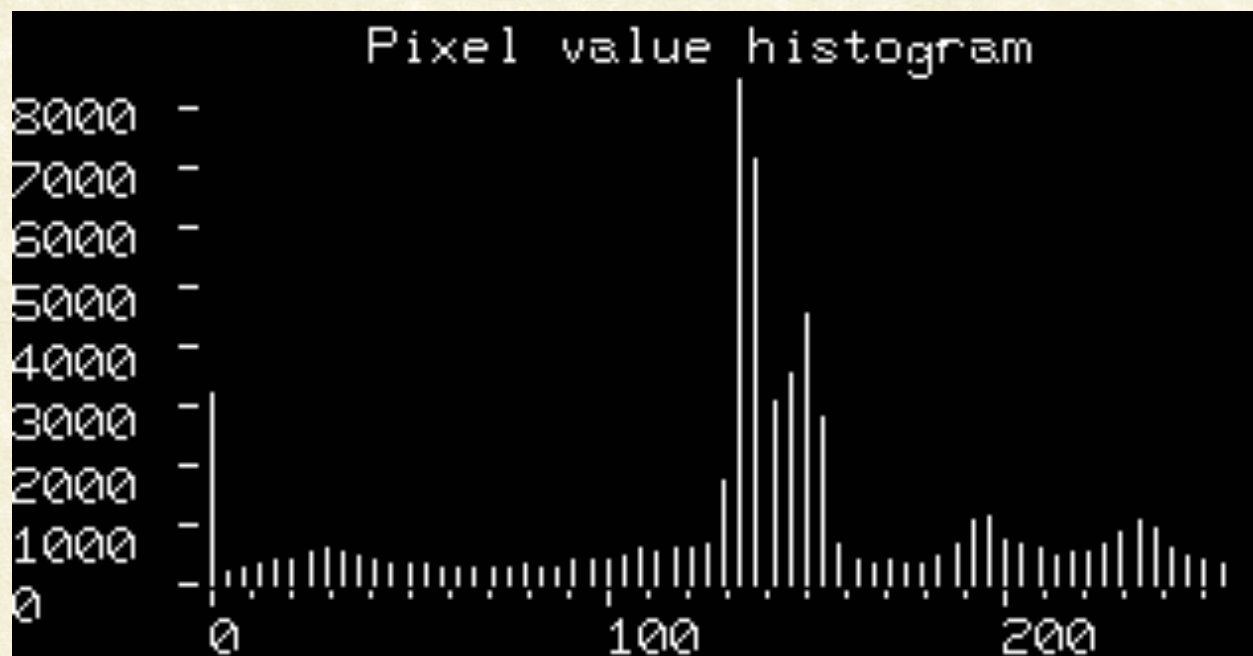
$$\hat{a}_{\text{low}} = \min\{i \mid H(i) \geq M \cdot N \cdot s_{\text{low}}\}$$

$$\hat{a}_{\text{high}} = \max\{i \mid H(i) \leq M \cdot N \cdot (1 - s_{\text{high}})\}$$

$$f_{\text{mac}}(a) = \begin{cases} a_{\text{min}} & \text{for } a \leq \hat{a}_{\text{low}} \\ a_{\text{min}} + (a - \hat{a}_{\text{low}}) \cdot \frac{a_{\text{max}} - a_{\text{min}}}{\hat{a}_{\text{high}} - \hat{a}_{\text{low}}} & \text{for } \hat{a}_{\text{low}} < a < \hat{a}_{\text{high}} \\ a_{\text{max}} & \text{for } a \geq \hat{a}_{\text{high}} \end{cases}$$



Ver.-2

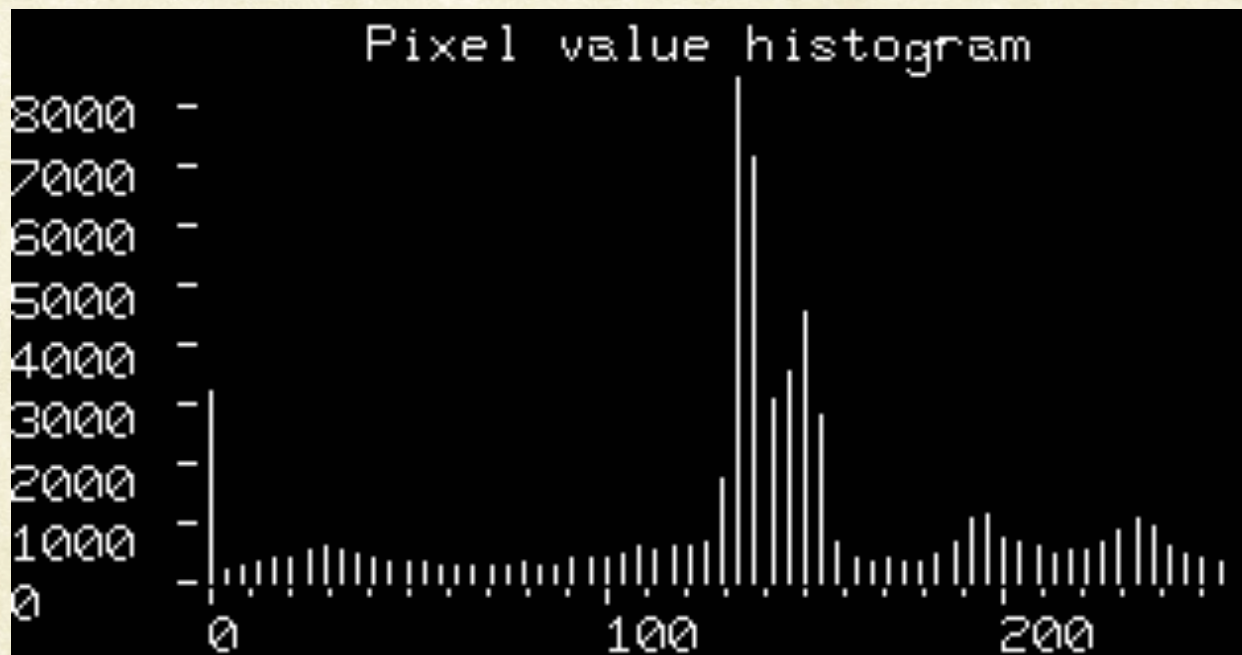




Are all intensities
Well-represented ?

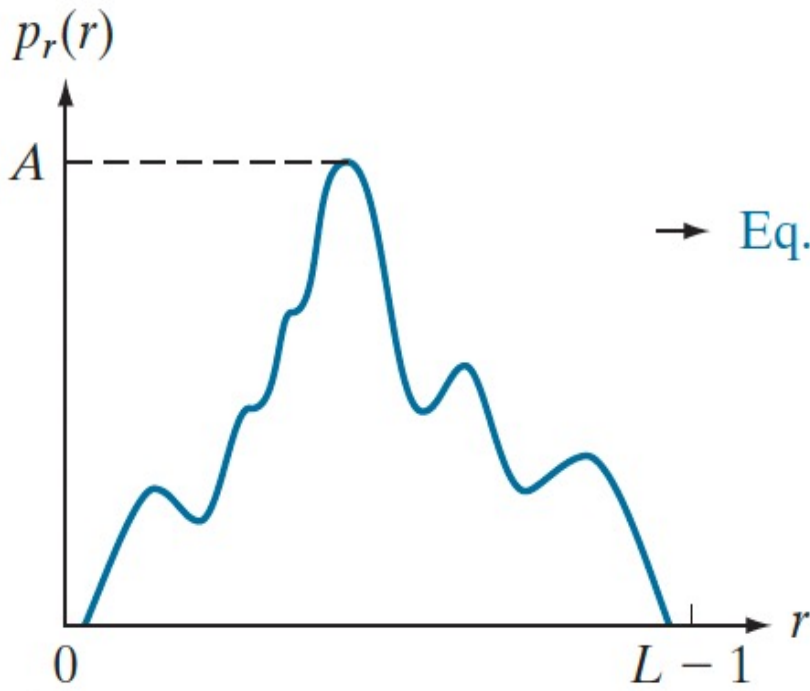
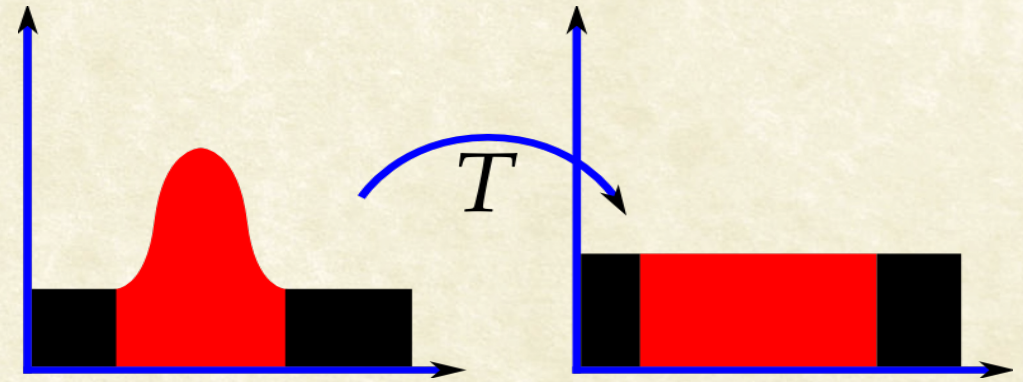
All Intensities
Matter

Ver. 2

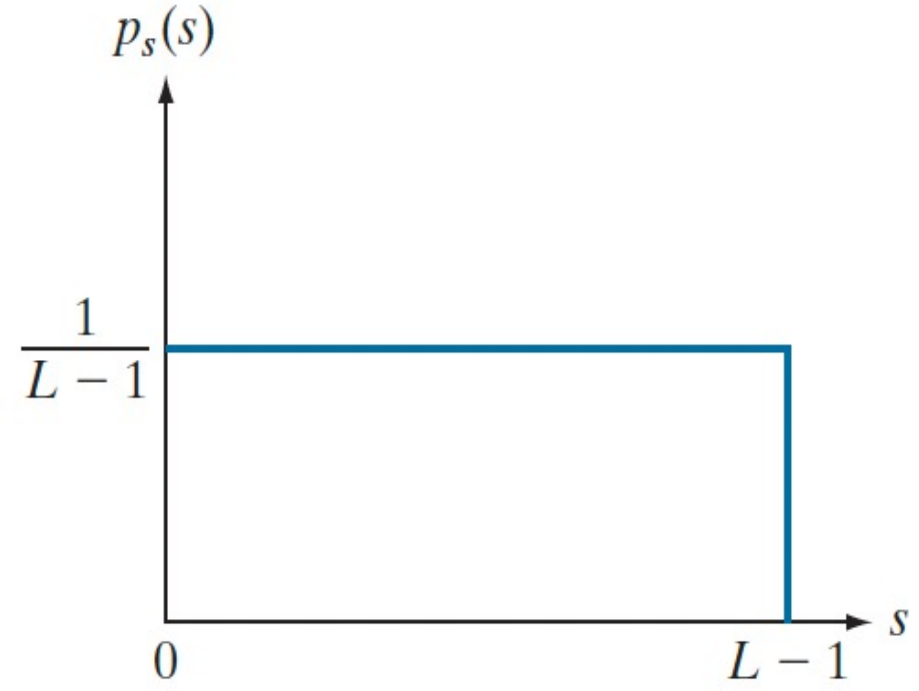




Histogram Equalization

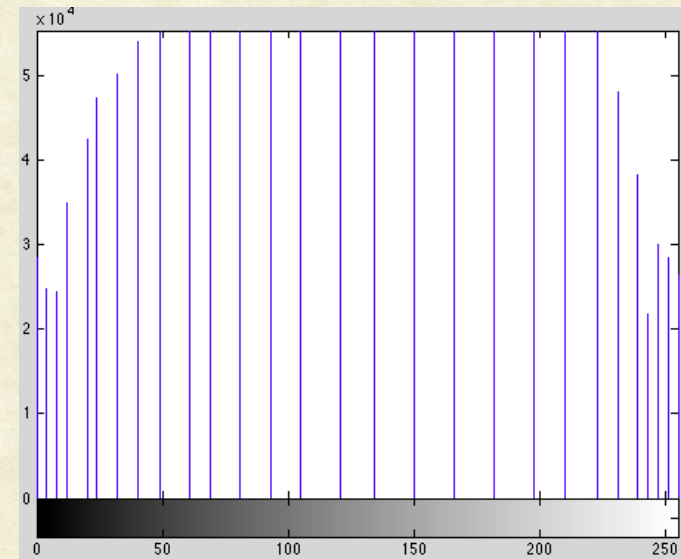
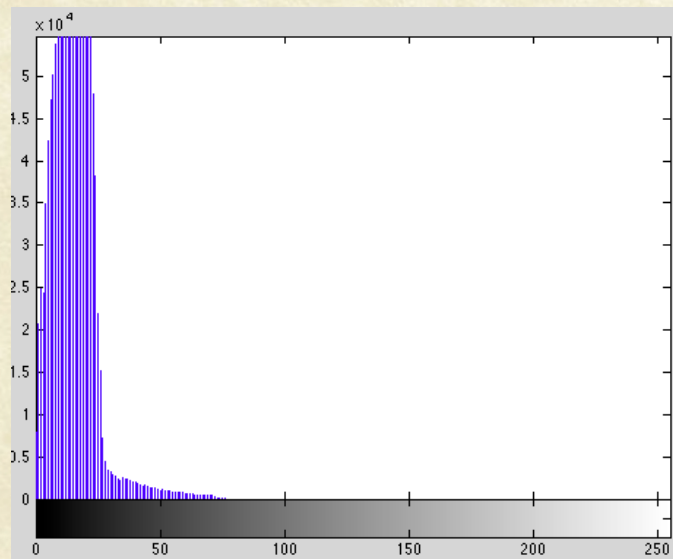
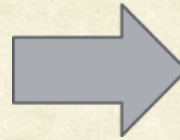


→ Eq. (3-11) →



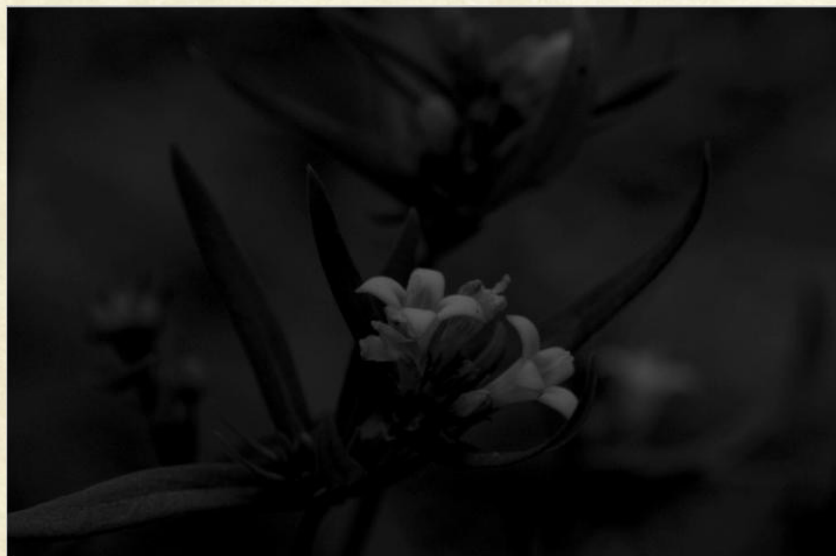


Histogram Equalization





Histogram Equalization



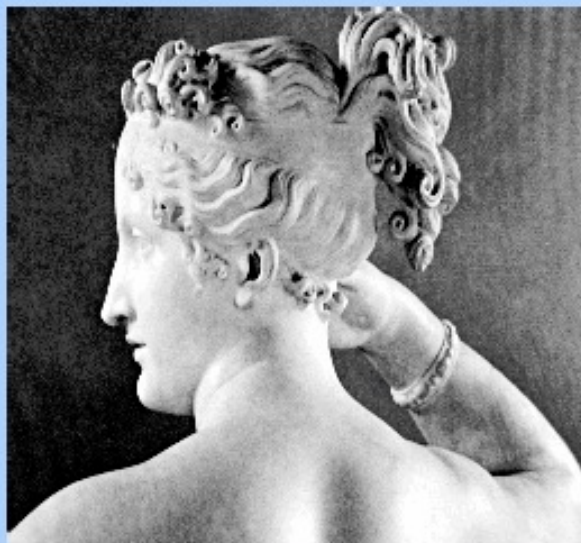
Contrast
Stretching



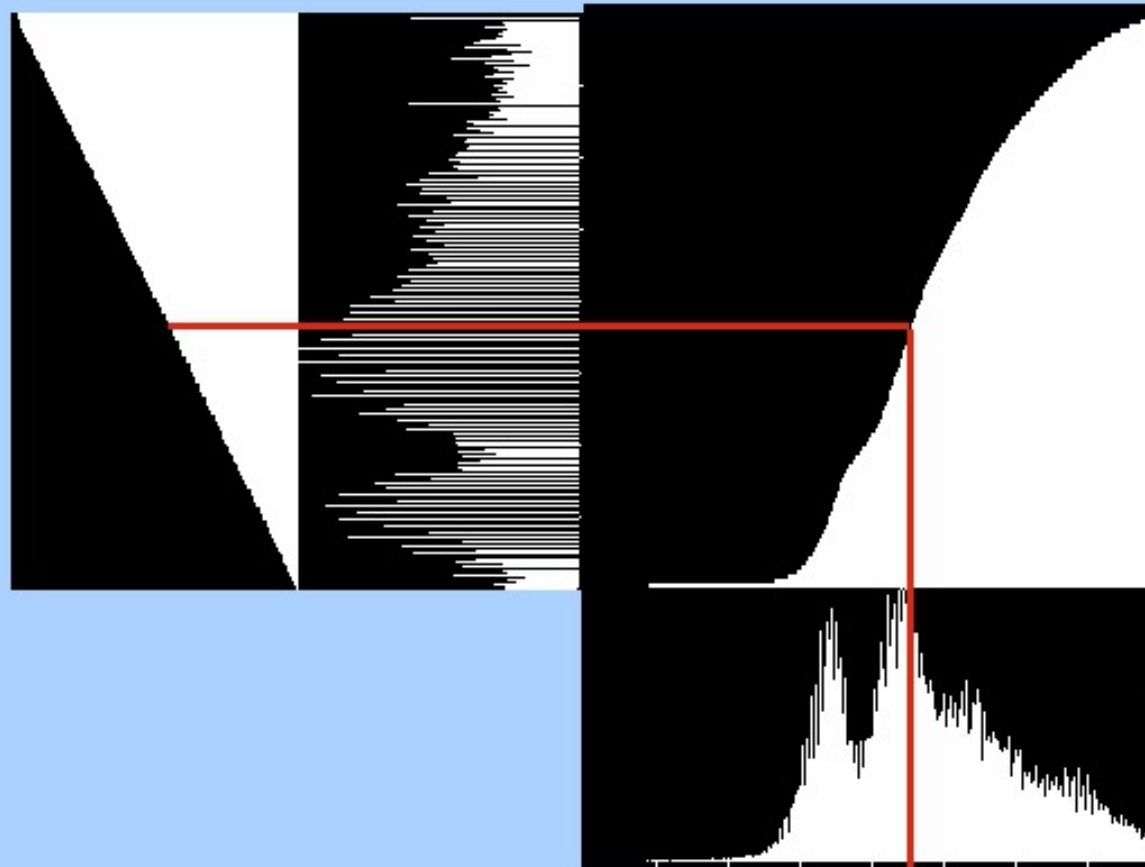
Histogram
Equalization



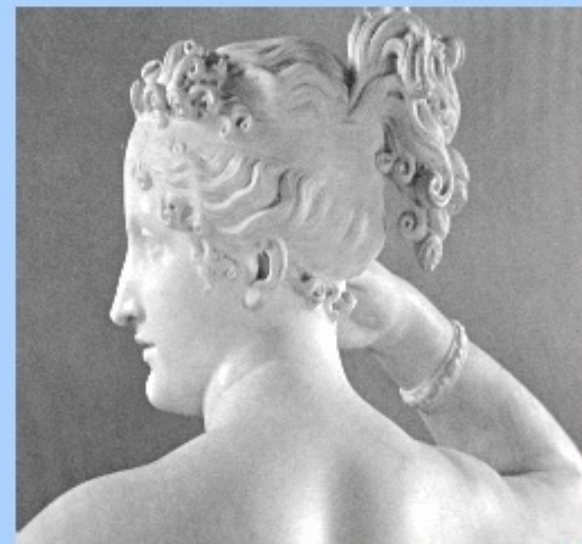
Computing the Intensity-Mapping



Equalized histogram



Histogram of original image

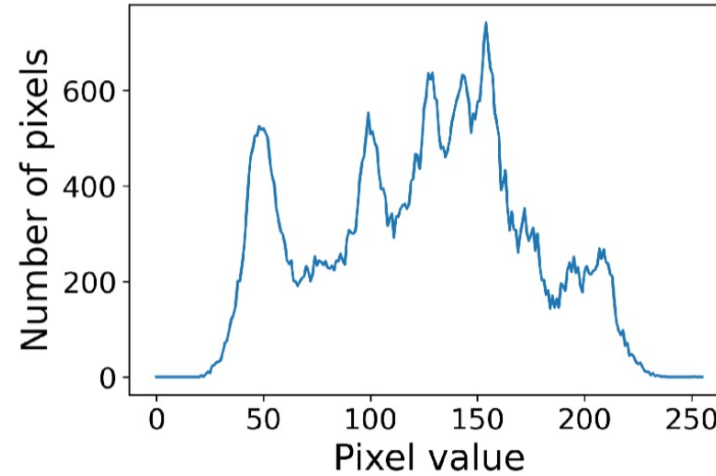




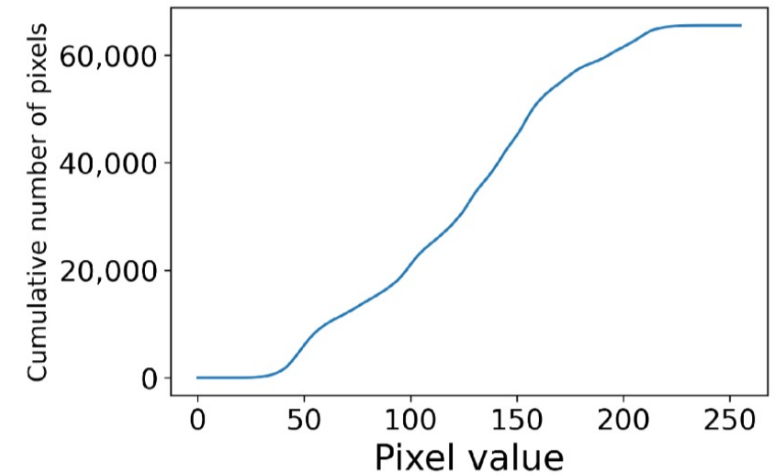
Computing the Mapping for Histogram Eq.



(a) Original image



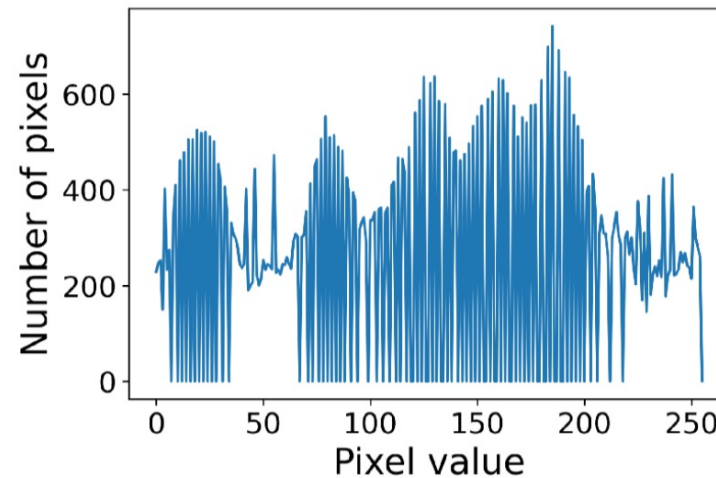
(b) Histogram



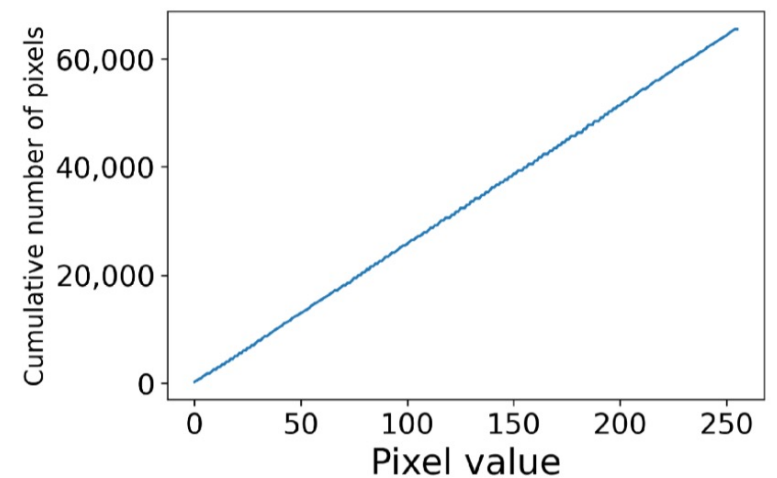
(c) Cumulative histogram



(d) Histogram-equalized image



(e) Histogram



(f) Cumulative histogram



Histogram Equalization

$$p_r(r_k) = \frac{n_k}{MN}$$

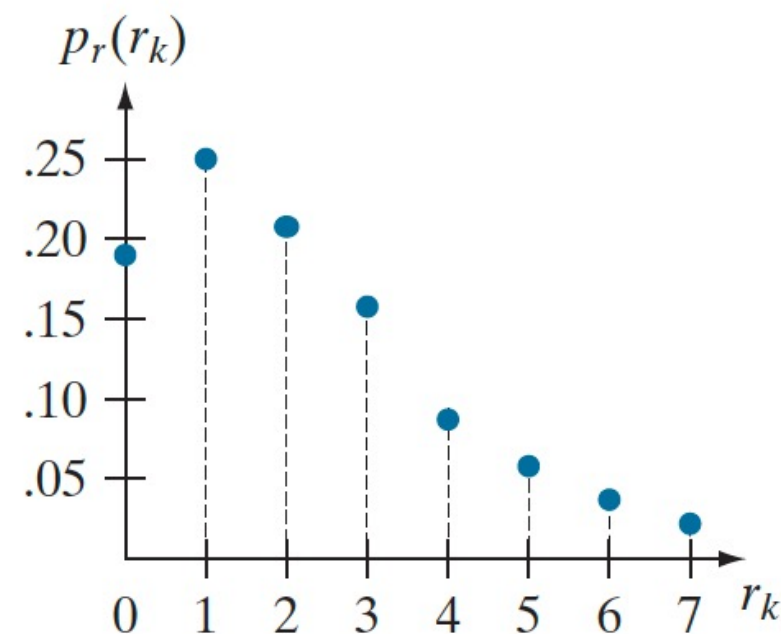
$$s_k = T(r_k) = \text{round} \left[(L - 1) \sum_{j=0}^k p_r(r_j) \right]; \quad k = 0, 1, 2, \dots, L - 1$$



Histogram Equalization - Example

r_k	n_k	$p_r(r_k) = n_k / MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

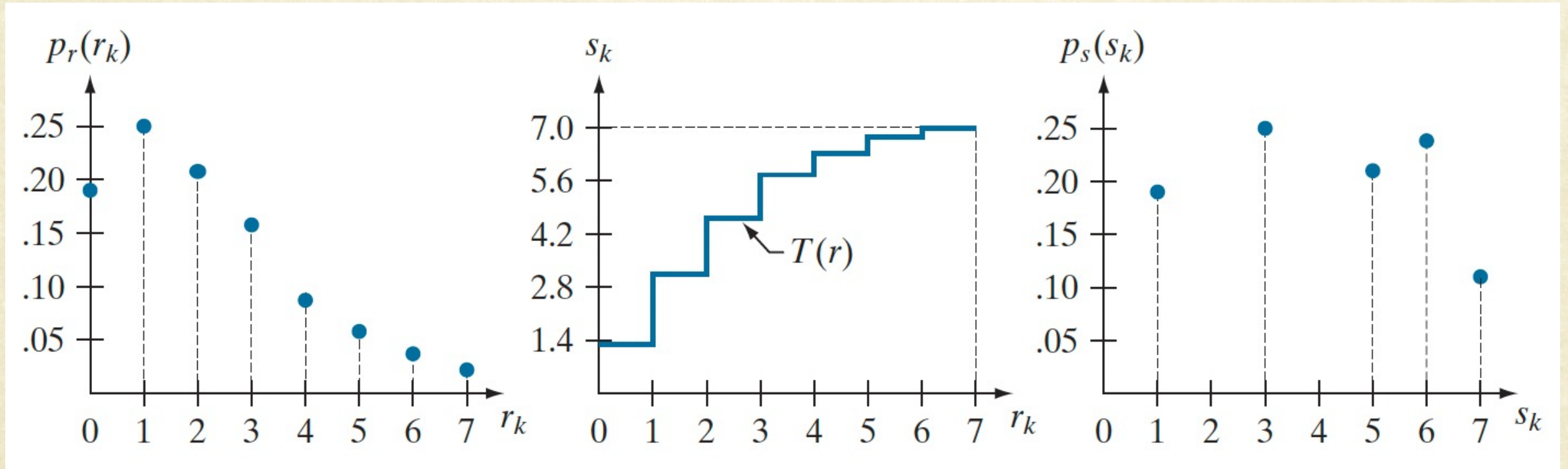
Intensity distribution and histogram values for a 3-bit, 64 x 64 digital image.





Histogram Equalization - Example

$$s_k = \text{round} \left[(L - 1) \sum_{j=0}^k p_r(r_k) \right]$$



(a) Original histogram;

(b) Transformation function;

(c) Equalized histogram.

FIGURE 3.19: Histogram equalization.



Histogram Equalization

1



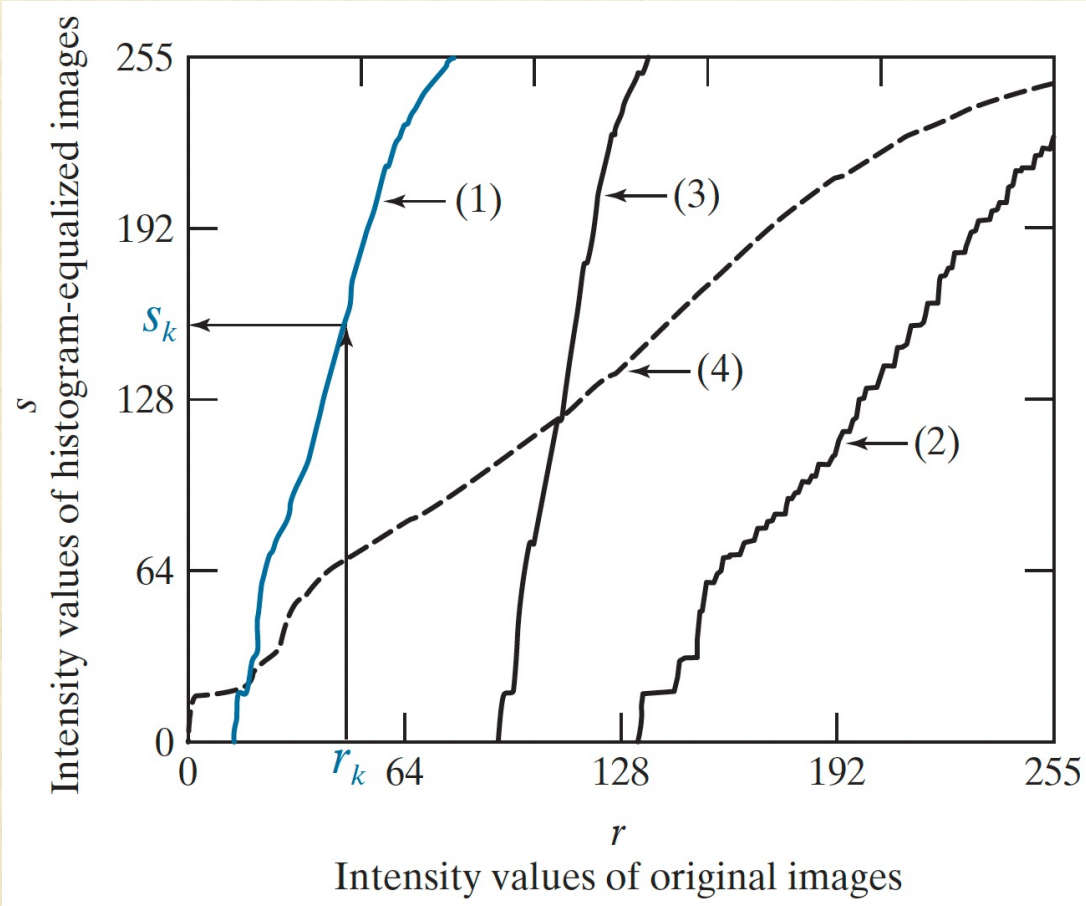
2



3



4





Histogram Equalization v/s Contrast Enhancement



Contrast Enhancement



Histogram equalization



Questions?